EPID 745 Epidemiologic Data Visualization

July 20th - 24th
1:30 PM - 5:00 PM
remotely via Zoom: https://jhjhm.zoom.us/my/erikwestlund

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Course Description: This course introduces students to the foundational principles and practical skills of data visualization, with a focus on both exploratory data analysis and enhancing the clarity and impact of statistical communication.

The course emphasizes visualization as a critical tool for statistical reasoning, model interpretation, and scientific communication. Students will learn to design and evaluate visualizations that are accurate, clear, and audience-appropriate. Topics include using visualization for exploratory data analysis, common pitfalls in data presentation, best practices for visual storytelling, and the role of graphics in conveying complex analytical insights.

Hands-on exercises will focus primarily on R, using ggplot2 and the tidyverse, with supplementary coverage of Python, Stata, and tools for interactive dashboards.

Course Materials: CTools website
We will use Kieran Healy's Data Visualization as a course text.

It is available for free online: https://socviz.co/. You can purchase a physical copy on Amazon if you wish.

Prerequisites: Experience with elementary statistics and statistical analysis software, ideally R.

Course Goals:

- Conceptualize and implement effective data visualizations using R and ggplot2.
- Learn common exploratory data analysis visualizations techniques that helps analysts understand their data and its structure
- Select appropriate visualization techniques based on data type, analytical goals, and audience needs.
- Critically evaluate data visualizations for effectiveness and accuracy.
- Communicate complex data insights through clear and impactful storytelling.

- Incorporate reproducibility and transparency into the data visualization process using modern coding practices.
- Address common challenges and pitfalls in data visualization, including choosing incorrect statistical transformations, misleading uses of scale and range, and poor design.
- Integrate visualizations into workflows for statistical modeling and applied data science.

Competencies: Use data visualization for improved statistical reasoning and communication.

Course Requirements:

Participation	30%
Problem Sets (3)	30% (10% each)
Final Project	40%
Total	100%

Classroom Expectations/Etiquette:

Attendance and participation in class sessions is required and formally assessed. Participation, both in discussions and applied practice, is an important part of the learning process. Please contact me if you must miss a session for an excused reason so we can plan a way to make up for missed class time.

Diversity, Equity, and Inclusion:

SPH is committed to creating classroom environments that are supportive of diversity, equity and inclusion.

Academic Integrity:

The faculty and staff of the School of Public Health believe that the conduct of a student registered or taking courses in the School should be consistent with that of a professional person. Courtesy, honesty, and respect should be shown by students toward faculty members, guest lecturers, administrative support staff, community partners, and fellow students. Similarly, students should expect faculty to treat them fairly, showing respect for their ideas and opinions and striving to help them achieve maximum benefits from their experience in the School.

Student academic misconduct refers to behavior that may include plagiarism, cheating, fabrication, falsification of records or official documents, intentional misuse of equipment or materials (including library materials), and aiding and abetting the perpetration of such acts. Please visit https://sph.umich.edu/student-resources/mph-mhsa.html for the full Policy on Student Academic Conduct Standards and Procedures.

SPH Writing Lab:

The SPH Writing Lab is located in 5025 SPH II and offers writing support to all SPH students for course papers, manuscripts, grant proposals, dissertations, personal statements, and all other academic writing tasks. The Lab can also help answer questions on academic integrity. To learn more or make an appointment, please visit the SPH writing lab website.

Student Well-Being:

SPH faculty and staff believe it is important to support the physical and emotional well-being of our students. If you have a physical or mental health issue that is affecting your performance or participation in any course, and/or if you need help connecting with University services, please contact the instructor or the SPH Office for Student Engagement and Practice. Please visit https://sph.umich.edu/student-life/wellness.html for information on wellness resources available to you.

Student Accommodations:

Students should speak with their instructors before or during the first week of classes regarding any special needs. Students can also visit the SPH Office for Student Engagement and Practice for assistance in coordinating communications around accommodations. Students seeking academic accommodations should register with Services for Students with Disabilities (SSD). SSD arranges reasonable and appropriate academic accommodations for students with disabilities. Please visit https://ssd.umich.edu/topic/our-services for more information on student accommodations.

Students who expect to miss classes, examinations, or other assignments as a consequence of their religious observance shall be provided with a reasonable alternative opportunity to complete such academic responsibilities. It is the obligation of students to provide faculty with reasonable notice of the dates of religious holidays on which they will be absent. Please visit http://www.provost.umich.edu/calendar/ for the complete University policy.

Course Topics/Reading List:

Date/Time/Topic	**Due**	**Notes & Exercises**
Day 1	Healy, chs. 1–3	- Why visualize? What makes a good visualization? - Intro to tools: R, Git, GitHub, editors, notebooks - Notebook formats: RMarkdown, Quarto, Jupyter (brief overview) - Core charts: bar charts, histograms, scatterplots - Levels of measurement and why they matter - Hands-on: set up notebook, load data, transform, plot, commit Homework: Problem Set 1 due Wednesday before class
Day 2		- The grammar of graphics (as a concept & as software) - Avoiding common ways visualizations mislead - Visualization as layers: data → aesthetics → geoms → stats → labels - Aggregation: built-in vs. manual - Working with facets and fills - Annotation and design principles (color, readability, clarity) - Hands-on: build a layered and well-labeled plot
Day 3	Cleveland reading online	- Data visualization for exploratory data analysis - Explore plots for studying location, spread, and

		structure of data Homework: Problem Set 2 due Thursday before class
Day 4	Healy, chs. 4–6 Problem set 1 due	Choosing what to visualize: causal questions and lazy DAGs - Generative experimentation - Plot types: time series, forest plots, ridgeline plots, maps Homework: Problem Set 3 due Friday at noon
Day 5	Healy, chs. 7–8 Problem set 2 due	- Extra time to catch up on what was missed - Dashboards: brief intro to Shiny and alternatives (demo or scaffolded build) - Hands-on: extend a basic Shiny app or explore an existing dashboard
End of week	Problem set 3 due at noon	
Later before grades submission	Final project due at 11:59 pm	

APPENDIX: CEPH-required Learning Objectives and Competencies *[Please list the learning objectives and competencies relevant to your course in the Course Goals and Competencies sections. CEPH expects that all listed competencies are formally assessed by the instructor, so please only choose those that are assessed through course assignments. Delete the appendix from your syllabus when you are done with it.* ***This applies to graduate-level courses only.****]*

Foundational Learning Objectives

Profession & Science of Public Health

1. Explain the role of quantitative and qualitative methods and sciences in describing and assessing a population's health
2. Explain the critical importance of evidence in advancing public health knowledge

Foundational Competencies

Evidence-based Approaches to Public Health

1. Apply epidemiological methods to the breadth of settings and situations in public health practice
2. Select quantitative and qualitative data collection methods appropriate for a given public health context
3. Analyze quantitative and qualitative data using biostatistics, informatics, computer-based programming and software, as appropriate
4. Interpret results of data analysis for public health research, policy or practice

Communication

5. Select communication strategies for different audiences and sectors
6. Communicate audience-appropriate public health content, both in writing and through oral presentation